StockSense Telemetry System

Costed Bill of Materials (BOM) — Hardware Revision v1.0

Date: 04th November 2026
Currency: LKR (Sri Lankan Rupee)

ACTIVE COMPONENT TOTAL	PASSIVES & HARDWARE	EST. BOARD UNIT COST
LKR 3,670.00	**LKR 501.00**	**LKR 4,971.00**
Microcontroller, Sensors & ICs	Power, Diodes, Terminal Blocks	Includes 2-Layer Custom PCB

1. CORE MICROCONTROLLER & SENSOR MODULES

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT (LKR)	TOTAL (LKR)
U2	ESP32-WROOM-32U Microcontroller Module (External IPEX Antenna)	Castellated38	1	1,850.00	1,850.00
J5 (HX711)	HX711 24-Bit ADC Weighing Sensor Module	Header 1x04 (2.54mm)	2	250.00	500.00
J6 (SHT30)	SHT30 I2C Temperature & Humidity Sensor Breakout	Header 1x04 (2.54mm)	1	600.00	600.00
J7 (RC522)	RFID-RC522 13.56MHz SPI Reader Module Kit	Header 1x08 (2.54mm)	1	320.00	320.00
J3 (Mini360)	Mini360 MP2307 Synchronous Buck Converter Module (12V to 5V)	Castellated 17x11mm	1	350.00	350.00
U1	AP2112K-3.3TRG1 LDO Regulator 3.3V 600mA	SOT-23-5	1	50.00	50.00

2. POWER MANAGEMENT & PROTECTION DISCRETES

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT (LKR)	TOTAL (LKR)
Q1	AO3401A P-Channel MOSFET (-30V, -4.2A, 44mΩ)	SOT-23-3	1	35.00	35.00
D2	12V 0.5W Zener Diode (BZX84C12L Gate Clamp)	SOT-23 / SOD-123	1	15.00	15.00
D1	SMAJ15A TVS Diode 15V Unidirectional (Transient Suppression)	DO-214AC (SMA)	1	45.00	45.00
D_BAT1, D_MAINS1	SS34 Schottky Barrier Diode (40V, 3A, Low Vf)	DO-214AB (SMC)	2	30.00	60.00

3. PASSIVE COMPONENTS, CONNECTORS & BOARD COST

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT (LKR)	TOTAL (LKR)
R1, R2, R3	10kΩ 1% 0805 SMD Resistors	R0805	3	5.00	15.00
R_DIV1	30kΩ 1% 0805 SMD Resistor (Mains Sense High)	R0805	1	5.00	5.00
R4, R5	4.7kΩ 1% 0805 SMD Resistors (I2C Pull-ups)	R0805	2	5.00	10.00
C1,C5,C7,C9..11	0.1µF 50V Ceramic Capacitors X7R 0805	C0805	6	8.00	48.00
C3, C4, C6, C8	1µF / 4.7µF / 10µF Ceramic Decoupling Capacitors	C0805 / C1206	4	12.00	48.00
C2	100µF 25V Electrolytic Capacitor (Low ESR)	Radial / SMD 6.3mm	1	35.00	35.00
J1	DC Barrel Power Jack (2.1mm x 5.5mm PCB Mount)	BarrelJack_Horizontal	1	85.00	85.00
J2	2-Pin Screw Terminal Block (5.08mm Pitch, Battery In)	TerminalBlock_5.08mm	1	60.00	60.00
SW1, SW2	Tactile Push Button Switches (6x6mm EN & Boot)	SW_PUSH_6mm	2	20.00	40.00
PCB Fabrication	Custom 2-Layer FR4 PCB (Est. per unit cost on batch of 5)	100x80mm Green HASL	1	800.00	800.00

TOTAL ESTIMATED BOM & BOARD FABRICATION COST (LKR)	LKR 4,971.00

Local Sri Lanka Procurement Notes:

- Prices based on prevailing retail rates from local electronics distributors (Tronic.lk, Senith Electronics, Alphatronic, Microchip.lk).
- Passive component prices (0805 resistors & capacitors) are estimated based on cut-tape purchases in small quantities.
- PCB Fabrication per-unit cost assumes a standard 5-board prototype run via JLCPCB/PCBWay shipped to Sri Lanka.